Derivation of the Vlasov Equation from N-Particle Hamiltonian Dynamics

Joseph K. Miller

May 23, 2026

1 Setup

Fix a dimension $d \geq 1$ and a pair potential $W : \mathbb{R}^d \to \mathbb{R}$. We consider N identical particles of unit mass in $\mathbb{R}^d$ with positions $x_i(t) \in \mathbb{R}^d$ and velocities $v_i(t) \in \mathbb{R}^d$, interacting through W in the mean-field scaling. The Hamiltonian on phase space $(\mathbb{R}^d \times \mathbb{R}^d)^N$ is

$$H_N(\mathbf{X}, \mathbf{V}) \;=\; \sum_{i=1}^{N} \frac{|v_i|^2}{2} \;+\; \frac{1}{N} \sum_{1 \leq i < j \leq N} W(x_i - x_j), \tag{1}$$

where $\mathbf{X} = (x_1, \ldots, x_N)$ and $\mathbf{V} = (v_1, \ldots, v_N)$. The factor $1/N$ is the mean-field normalization, chosen so that the kinetic and interaction energies are of the same order as $N \to \infty$.

The corresponding Hamilton equations are

$$\dot{x}_i \;=\; v_i, \qquad \dot{v}_i \;=\; -\frac{1}{N} \sum_{j \neq i} \nabla W(x_i - x_j), \qquad i = 1, \ldots, N. \tag{2}$$

Assumption 1. The potential $W \in C^{1,1}(\mathbb{R}^d)$ is even, $W(-x) = W(x)$, with globally Lipschitz gradient:

$$L \;:=\; \operatorname{Lip}(\nabla W) \;=\; \sup_{x \neq y} \frac{|\nabla W(x) - \nabla W(y)|}{|x - y|} \;<\; \infty.$$

Under Assumption 1, the system (2) admits a unique global C^1 solution for every initial datum $(\mathbf{X}_0, \mathbf{V}_0)$, and the symmetric group S_N acts on the solution by permutation of indices.

2 The Empirical Measure

Definition 2. The *empirical measure* associated to a configuration $\mathbf{Z} = (\mathbf{X}, \mathbf{V}) \in (\mathbb{R}^d \times \mathbb{R}^d)^N$ is the probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ defined by

$$\mu^N[\mathbf{Z}] \;:=\; \frac{1}{N} \sum_{i=1}^{N} \delta_{(x_i, v_i)}.$$

For a solution $t \mapsto (\mathbf{X}(t), \mathbf{V}(t))$ of (2), we write $\mu^N_t := \mu^N[\mathbf{X}(t), \mathbf{V}(t)]$.

Proposition 3 (Weak evolution of the empirical measure)**.** *Let* $(\mathbf{X}, \mathbf{V}) : [0,T] \to (\mathbb{R}^d \times \mathbb{R}^d)^N$ *solve* (2)*, and let* $\rho_t^N(x) := \int_{\mathbb{R}^d} \mu_t^N(dx, dv)$ *denote the spatial marginal. Then for every* $\varphi \in C_c^\infty(\mathbb{R}^d \times \mathbb{R}^d)$*,*

$$\frac{d}{dt}\langle \mu_t^N, \varphi\rangle \;=\; \big\langle \mu_t^N,\; v\cdot\nabla_x\varphi - (\nabla W * \rho_t^N)\cdot\nabla_v\varphi\big\rangle \;+\; R_N(t), \tag{3}$$

where the remainder satisfies $|R_N(t)| \le \frac{1}{N}\|\nabla W\|_\infty \|\nabla_v\varphi\|_\infty$ *and vanishes identically when* $\nabla W(0) = 0$ *(in particular, under Assumption 1).*

Proof. Differentiating $\langle \mu_t^N, \varphi\rangle = \frac{1}{N}\sum_i \varphi(x_i, v_i)$ in t and using (2),

$$\frac{d}{dt}\langle \mu_t^N, \varphi\rangle \;=\; \frac{1}{N}\sum_{i=1}^N v_i\cdot\nabla_x\varphi(x_i,v_i) \;-\; \frac{1}{N^2}\sum_{i=1}^N\sum_{j\neq i}\nabla W(x_i - x_j)\cdot\nabla_v\varphi(x_i,v_i).$$

The first term equals $\langle \mu_t^N, v\cdot\nabla_x\varphi\rangle$. For the second, add and subtract the diagonal $i = j$:

$$\frac{1}{N^2}\sum_{i,j}\nabla W(x_i - x_j)\cdot\nabla_v\varphi(x_i,v_i) \;-\; \frac{1}{N^2}\sum_i \nabla W(0)\cdot\nabla_v\varphi(x_i,v_i).$$

The double sum is exactly $\int (\nabla W * \rho_t^N)(x)\cdot\nabla_v\varphi(x,v)\,d\mu_t^N(x,v)$, since $\rho_t^N = \frac{1}{N}\sum_i \delta_{x_i}$. Under Assumption 1 the potential W is even, hence $\nabla W(0) = 0$ and the diagonal term vanishes. In general it is bounded by $\frac{1}{N}\|\nabla W\|_\infty\|\nabla_v\varphi\|_\infty$. □

Corollary 4. *Under Assumption 1, the empirical measure* μ_t^N *is a (distributional) solution to the nonlinear Vlasov equation* (4) *below, in the sense that it satisfies* (3) *with* $R_N \equiv 0$ *for every* $\varphi \in C_c^\infty(\mathbb{R}^d \times \mathbb{R}^d)$*.*

3 The Vlasov Equation

Formally passing $N \to \infty$ in (3) and assuming $\mu_t^N \to f_t$ weakly, we obtain the limit equation

$$\boxed{\partial_t f \;+\; v\cdot\nabla_x f \;-\; (\nabla W * \rho_t)(x)\cdot\nabla_v f \;=\; 0, \qquad \rho_t(x) = \int_{\mathbb{R}^d} f(t,x,v)\,dv.} \tag{4}$$

This is the *nonlinear Vlasov equation* associated to the pair potential W. We collect the basic well-posedness statement.

Theorem 5 (Existence and uniqueness for Vlasov)**.** *Let* $f_0 \in \mathcal{P}_1(\mathbb{R}^d \times \mathbb{R}^d)$ *be a probability measure with finite first moment. Under Assumption 1, there exists a unique narrowly continuous family* $t \mapsto f_t \in \mathcal{P}_1(\mathbb{R}^d \times \mathbb{R}^d)$ *satisfying* (4) *in the distributional sense with* $f_{|t=0} = f_0$*.*

Existence and uniqueness follow from the characteristic flow construction: define the mean-field ODE

$$\dot X(t,z) = V(t,z), \qquad \dot V(t,z) = -(\nabla W * \rho_t)(X(t,z)), \qquad (X,V)(0,z) = z, \tag{5}$$

with $\rho_t = (X(t,\cdot))_\#(\int f_0\,dv)$, and set $f_t = (X(t,\cdot), V(t,\cdot))_\# f_0$. The fixed-point argument for (5) uses the Lipschitz hypothesis on ∇W and the Wasserstein-1 metric on probability measures.

4 The Mean-Field Limit: Dobrushin's Theorem

The empirical-measure framework reduces the mean-field limit to a stability statement for measure-valued solutions of (4).

Theorem 6 (Dobrushin, 1979)**.** *Under Assumption 1, there exists a constant $C = C(L) > 0$ such that any two measure solutions $f_t, g_t \in \mathcal{P}_1(\mathbb{R}^d \times \mathbb{R}^d)$ of (4) satisfy*

$$W_1(f_t, g_t) \;\le\; e^{Ct}\, W_1(f_0, g_0), \qquad t \ge 0, \tag{6}$$

where W_1 denotes the Wasserstein-1 distance.

Corollary 7 (Mean-field limit)**.** *Let $f_0 \in \mathcal{P}_1(\mathbb{R}^d \times \mathbb{R}^d)$, and let $(\mathbf{X}_0^N, \mathbf{V}_0^N)$ be initial data with empirical measure μ_0^N satisfying $W_1(\mu_0^N, f_0) \to 0$ as $N \to \infty$. Then, under Assumption 1, for every $T > 0$,*

$$\sup_{t\in[0,T]} W_1(\mu_t^N, f_t) \;\le\; e^{CT} W_1(\mu_0^N, f_0) \xrightarrow[N\to\infty]{} 0.$$

Proof sketch of Theorem 6. Let π_0 be an optimal coupling of f_0, g_0 for W_1. Let $(X_f, V_f), (X_g, V_g)$ denote the characteristic flows (5) associated to f and g, respectively, and define $\pi_t := (X_f(t,\cdot), V_f(t,\cdot), X_g(t,\cdot), V_g(t$ viewed as a coupling of f_t and g_t. Then

$$W_1(f_t, g_t) \;\le\; \int |X_f - X_g| + |V_f - V_g|\, d\pi_0.$$

A Gronwall argument on the characteristic equations, using $\mathrm{Lip}(\nabla W) = L$ and the bound $|\nabla W * \rho - \nabla W * \sigma|_\infty \le L\, W_1(\rho, \sigma)$, gives the exponential estimate (6). □

5 Hamiltonian Structure of the Limit

The Vlasov equation (4) inherits a Hamiltonian structure from the N-particle system (1). On the space of densities $f \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d)$, define the energy

$$\mathcal{H}[f] \;:=\; \int \frac{|v|^2}{2} f\, dx\, dv \;+\; \frac{1}{2} \iint W(x - x') f(x,v) f(x',v')\, dx\, dv\, dx'\, dv',$$

and the Lie–Poisson bracket

$$\{F, G\}[f] \;:=\; \int f \left\{ \tfrac{\delta F}{\delta f},\, \tfrac{\delta G}{\delta f} \right\}_{\mathrm{can}} dx\, dv,$$

where $\{\cdot,\cdot\}_{\mathrm{can}}$ is the canonical Poisson bracket on $T^*\mathbb{R}^d$. Equation (4) is then equivalent to

$$\partial_t f \;=\; \{f, \mathcal{H}\}.$$

A rigorous treatment of this structure—identifying the bracket as the Lie–Poisson bracket on the dual of the Lie algebra of Hamiltonian vector fields and showing that it arises as the $N \to \infty$ limit of the canonical brackets on $(\mathbb{R}^d \times \mathbb{R}^d)^N$—is the subject of [4].

References

[1] R. L. Dobrushin, *Vlasov equations*, Funct. Anal. Appl. **13** (1979), 115–123.

[2] H. Neunzert, *An introduction to the nonlinear Boltzmann–Vlasov equation*, in: Kinetic theories and the Boltzmann equation (Montecatini, 1981), Lecture Notes in Math. **1048**, Springer, 1984.

[3] H. Spohn, *Large Scale Dynamics of Interacting Particles*, Springer, 1991.

[4] J. K. Miller, A. R. Nahmod, N. Pavlović, M. Rosenzweig, G. Staffilani, *A Rigorous Derivation of the Hamiltonian Structure for the Vlasov Equation*, Forum of Mathematics, Sigma **11** (2023), e77.

[5] F. Golse, *On the dynamics of large particle systems in the mean field limit*, in: Macroscopic and large scale phenomena: coarse graining, mean field limits and ergodicity, Lect. Notes Appl. Math. Mech. **3**, Springer, 2016, 1–144.